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Madhapur, Hyderabad, Telangana – 500081

WEIGHT GENE GUIDE

Comprehensive Obesity Genetic Risk Screening

Warren Alemao

PATIENT INFORMATION

Patient Name: Warren Alemao

Age: 43 **Gender:** Male

Sample Details:

Collection Date	20 Nov 2024
Type of sample	Buccal swab
Genomic Specimen ID	KHDEOMGPCSP17

Sequencing Details:

Sequencing Type	Germline
WGS/WES/Targeted Seq	Whole Exome Sequencing
Encoding	Illumina 1.9
Sequence length	151bp
Overall Alignment rate	99.85%
Mean Sequencing depth	190.0
Q30 score	93.85%

INDICATION FOR TESTING

Warren Alemao, **43**, **Male** was referred for genetic testing to analyze risk for Obesity weight gain.

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WELCOME

Date: 06 Dec 2024

Dear Warren Alemao

Welcome to Weight Gene Guide, a comprehensive and personalized genetic analysis that utilizes advanced DNA sequencing technologies, including Next-Generation Sequencing (NGS). This analysis is designed to provide you and your healthcare team with valuable insights into your genetic predisposition to obesity. Our in-depth DNA data analytics reveal how your unique genetic profile influences critical aspects such as your body's response to diet, metabolism, and lifestyle.

These insights allow us to provide you with tailored recommendations to manage and potentially mitigate your risk of obesity. At Weight Gene Guide, we are committed to supporting your journey to better health. Your well-being is at the core of everything we do, and we take pride in delivering a data-driven approach to your care. Our goal is to empower you with the knowledge and tools you need to make informed decisions, guiding you toward long-term optimal health through personalized, genetics-driven strategies. Thank you for entrusting Weight Gene Guide with your health. We are here to support you every step of the way.

Warm Regards

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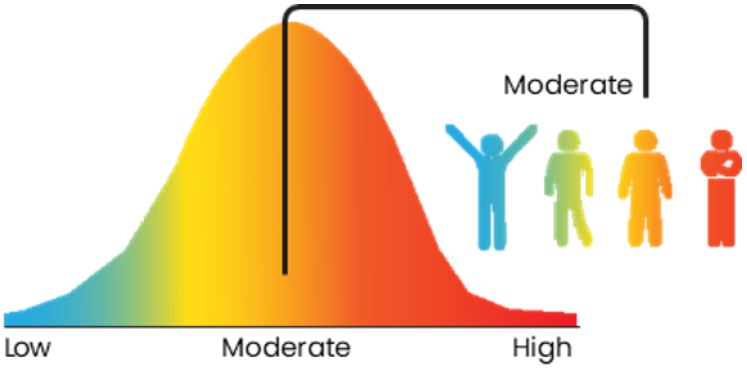
YOUR "WEIGHT-GENE" SUMMARY

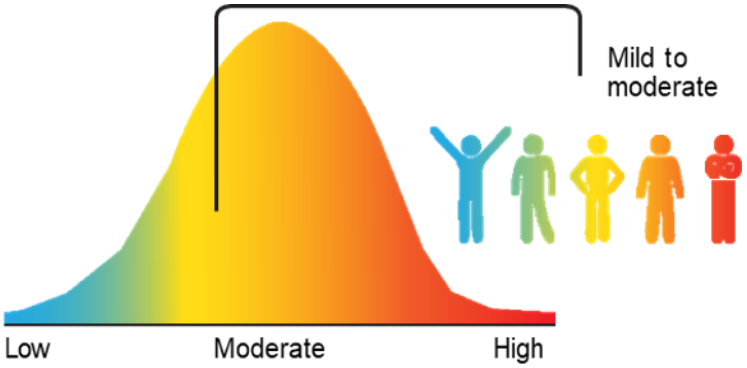
Obesity Risk

Genes Analyzed: 110
Variants analyzed: 4320



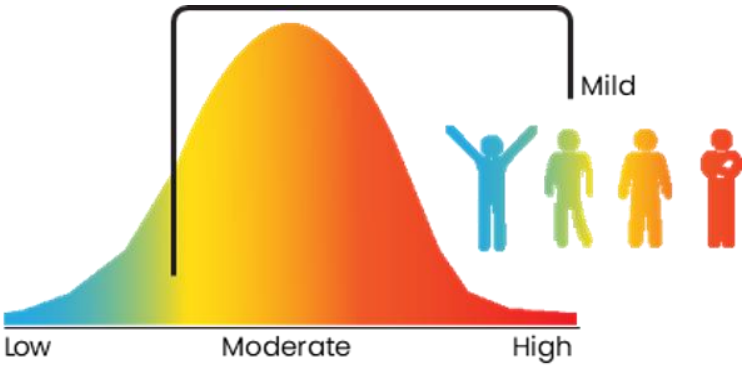
Insulin Sensitivity

Genes Analyzed: 98
Variants analyzed: 3760



Exercise Intolerance

Genes Analyzed: 141
Variants analyzed: 2746



Findings are statistically significant at a 95% confidence level

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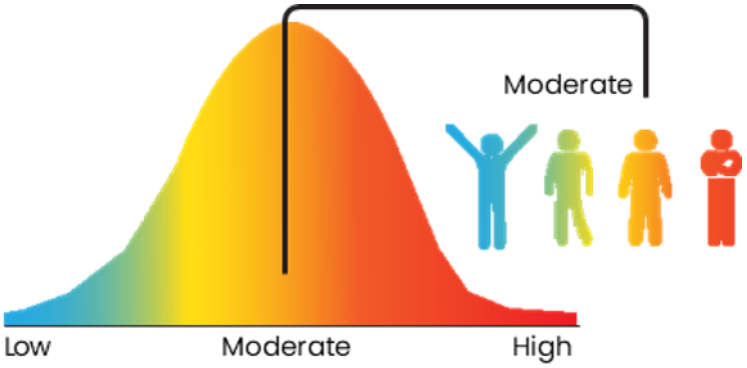
GENE-IUS EATING – YOUR NUTRIGENOMICS

Gene-ius Eating offers a personalized approach to nutrition, aligning your diet with your genetic makeup for optimal well-being and balanced health. It helps identify potential health risks, allowing for preventative measures and enhanced physical and cognitive performance.

Your Carbohydrate, Fat and Protein Gene Summary

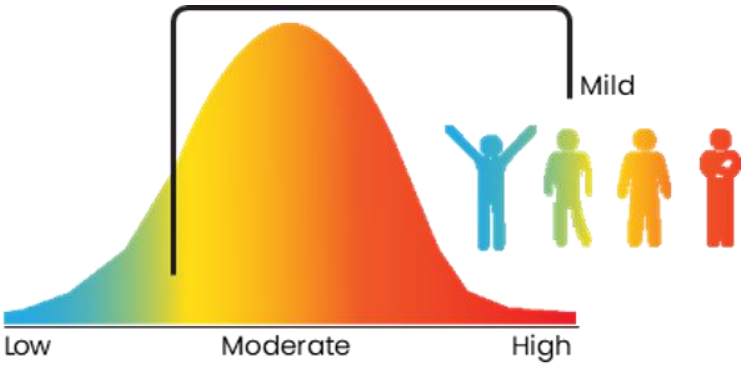
Carbohydrate Intolerance

Genes Analyzed: 115
Variants analyzed: 2838



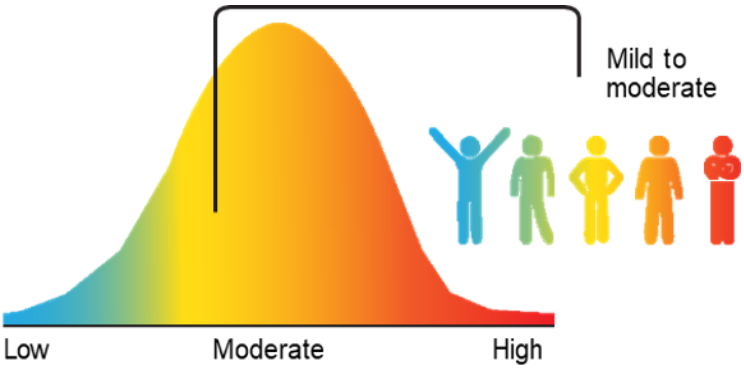
Fat Intolerance

Genes Analyzed: 79
Variants analyzed: 2122



Protein Intolerance

Genes Analyzed: 141
Variants analyzed: 2746



Findings are statistically significant at a 95% confidence level

DOCTOR'S RECOMMENDATIONS

- You have a **moderate** genetic risk for **obesity**. Clinically significant mutations are seen in LEPR gene, which are homozygous, along with some other pertinent genes.
- You have a **mild to moderate** risk for **insulin resistance**. Heterozygous mutations are seen in genes responsible for glucose metabolism, including HNF1A and HNF4A genes. These can cause both increased blood sugars as well as weight gain.
- You have a **mild** intolerance towards **exercise**. You are less likely to have any compliance issues with workouts. Hence, you can create a strong workout regimen including musculoskeletal strengthening exercises and cardio workouts.
- You have **moderate** intolerance towards **carbohydrates**. Several mutations are seen for impaired glucose metabolism, which can cause increased blood sugars as well as weight gain.
- You have **mild** intolerance for **fats**. Consumption of good quality fats is recommended. You are less likely to have any complications with KETO diets.
- You have **mild to moderate** intolerance for **proteins**. Avoid excessive intake of protein powders.

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Your Fitness and Nutrition Genes

Carbohydrate , Lipid , Protein, Musculoskeletal Metabolism

MUTATIONS SEEN IN GENES	RECOMMENDATIONS
ACSF3, CD320, CUBN, MTR, MTRR, MVK, TCN2	• Include dark green leafy vegetables, Vitamin B12 rich foods (Eggs, Fish, seafood). • Recommend Vitamin B12 and folate supplementation. Include dark green leafy vegetables, Vitamin B12 rich foods (Eggs, Fish, seafood).
COQ2, COQ4, COQ6, COQ7	• Include foods like oranges, strawberries, soybeans, lentils, peanuts.
NLRP12	• Avoid excessive cold exposure, ingestion of cold foods. Avoid excessive stress.
LAMA2, MUT	• Avoid heavy workouts. Please ensure to include foods like Pumpkin seeds, Walnuts, Peanuts, Almonds, Eggs, Buttermilk.
CDH23	• Recommend yearly eye exam and hearing tests after age 50
ARSA	• Include foods with complex carbohydrates. Fats to constitute only 20% of the total meal.
CRACR2B	• Avoid high fat diets and prolonged periods of fasting. Include riboflavin supplementation or riboflavin rich foods like eggs, dairy, green vegetables, fish.
PEX16, PEX2	• Can cause neuropathy, tingling, numbness, shooting pains. Include Fish Oil supplements.
ALDH5A1	• Avoid high protein diets. Include diet rich in complex carbohydrates.
FLVCR1	• Include foods which improve iron stores like dates, legumes, dark green vegetables. Poor oral absorption of iron seen with this mutation. Recommend monitoring CBC and Iron profile yearly.

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Your Fitness and Nutrition Genes

Carbohydrate , Lipid , Protein, Musculoskeletal Metabolism

MUTATIONS SEEN IN GENES	RECOMMENDATIONS
BCKDHA	• Avoid high protein meals/ powders. Increase intake of thiamine rich foods, like whole wheat, unpolished rice, millets.
KCNE1, RBM20, TIMM50	• Prefer exercises like brisk walking, strength training with lesser weights. Avoid Fasting
ENO3, GAA	• Need a pre workout snack rich in complex carbohydrates as glucose may not be readily available during work outs. Ideas for snacks include a banana, idli with peanut chutney, pomegranate, Avocado toast, oatmeal, almonds, walnuts. Make sure to eat the snack at least 45 minutes before work out.
COG4, COG6, DPM2, SLC16A1	• After work out, include a meal rich in complex carbohydrates and proteins.
IFITM5, VDR	• Include phosphate rich foods like sunflower seeds, pumpkin seeds, buttermilk, aged cheese, green leafy vegetables, beans
COL2A1, COL6A1, COL6A2, COL6A3, NEB	• To avoid muscle cramps after strenuous workouts, include foods like bone broth, nuts, beans, fish, broccoli, whole grains.
TRPM6	• Recommend daily magnesium supplementation
CDH23	• Milk, fresh cheeses, fresh milk cream can cause symptoms of bloating, malabsorption. Avoid fresh dairy. Can include buttermilk, aged cheeses. Increase consumption of fruits.

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Your Gene guide for healthy living

Avoid

- Heavy or excessive workouts and fasting.
- High-protein diets or meals/powders and Excessive stress.
- High-fat diets and prolonged periods of fasting.
- Fresh dairy products like milk, fresh cheeses, and cream (can cause bloating).
- Excessive cold exposure and ingestion of cold foods.

Include

- Dark green leafy vegetables, Vitamin B12-rich foods (eggs, fish, seafood).
- Foods with riboflavin (eggs, dairy, green vegetables, fish).
- Foods rich in complex carbohydrates and iron stores, like dates, legumes, and unpolished rice.
- Post-workout meals with complex carbohydrates and proteins (e.g., oatmeal, avocado toast).
- Thiamine-rich foods (whole wheat, millets, unpolished rice).
- Phosphate-rich foods (sunflower seeds, pumpkin seeds, buttermilk, aged cheeses, beans).

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Your Vitamins Summary

Analysis For	SCORE	Supplements
Vitamin-B12	1 2 3 4	Essential
Vitamin-B9 Folic Acid	1 2 3 4	Essential
Vitamin-C	1 2 3 4	Essential
Vitamin-B1	1 2 3 4	Advised
Vitamin-B2	1 2 3 4	Optional
Vitamin-B6 Biotin	1 2 3 4	Essential
Vitamin-E	1 2 3 4	Optional
Vitamin-D	1 2 3 4	Advised
Vitamin-K	1 2 3 4	Optional

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Your Minerals Summary

Analysis For	SCALE	Supplements
Iron	1 2 3 4	Essential
Magnesium	1 2 3 4	Advised
Molybdenum	1 2 3 4	Optional
Phosphate	1 2 3 4	Optional
Calcium	1 2 3 4	Advised
Zinc	1 2 3 4	Essential

Consult your healthcare provider before use; individual results may vary.

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Your Gene guide for healthy living

Recommendations

Pre-workout snacks rich in complex carbohydrates (banana, idli with peanut chutney, almonds).

Yearly eye exams and hearing tests after age 50.

Vitamin B12 rich foods and folate supplementation.

Daily magnesium supplementation to prevent muscle cramps.

Monitoring CBC and iron profile yearly due to poor iron absorption.

Prefer exercises like brisk walking and strength training with lighter weights.

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TOP BIOMARKERS LISTED

CONDITION	BIOMARKERS
Obesity Risk	*FTO, PCSK1, MC4R, LEPR, LEP, FABP2, ENPP1, POMC, KSR2, BDNF, SH2B1, ADCY3, ADRB3, GHRL, CEP19, ABCC8, MYH9, SIM1, NTRK2*
Insulin Sensitivity	*HNF1A, HNF4A, HNF1B, IPF1, NEUROD1, INS, ENPP1, LEPR, PCSK1, GCK, AGPAT2, BSCL2, CAV1, PTRF, LMNA, PPARG, CIDEC, ZMPSTE24, AKT2*
Exercise Intolerance	*FBN1, GYG1, RYR1, COMT, 5-HTT, MATN3, SMAD6, SLC26A2, COMP, NLRP1, FGFR1, FRZB, VDR, IL17F, IL-6, BMP2*
Carbohydrate intolerance	*FBP1, FBP2, NEDD4L, PC, G6PC3, G6PC2, GCK, MPI, HMGCL, MANBA, ALG2, PMM2, GALNT3, ALG13, DOLK, DPAGT1, SLC35A2*
Fat intolerance	*LEP, MC4R, IL6, POMC, MAP4K4, PPARG, ACADVL, MTTP, APOB, GATA1, CAD, HADHA, PCSK9, APOE, ASS1, SLC25A13, ARSD*
Protein intolerance	*DBT, PPM1K, BCAT2, HPD, OTC, CPS1, AASS,ASS1, NAGS, ASL, TAT, SLC7A9, SLC3A1, SLC7A7, SLC6A19, ETHE1, SLC25A15, ACAT1, SLC26A2*
Vitamins	*AICDA,C4BPB, ABCD4, ACSF3, ACAD12, ACAD13, TPK1, SLC44A4, CPT2, CYB5R3, ACAT1, BTD, MTHFD1, ADA, SCAR-B1, VKORC1, ABCA1, GGCX*
Minerals	*PCBD1,SARS2, AGA, AP2S1, BSND, CACNA1S, BMP1, CA2, CLCN7, ACSF3, BBOF1, CD320, DMT1,SLC7A7, SLC39A8, XDH*

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EVIDENCE-BASED DIET PATTERN

This diet plan is designed based on the genomic insights from your genomic data. It focuses on optimizing health by strengthening your immune system, promoting weight management, and addressing risks for chronic conditions. These tailored guidelines empower you to make informed decisions for achieving and maintaining long-term well-being.

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NUTRITION RECOMMENDATIONS

Carbohydrates

Grains which can be included in diet- Include Thiamine rich foods like Hand pounded rice, Single polished rice, Whole wheat flour, Millets (any kind), Quinoa.

- Hand pounded rice, basmati rice, whole wheat. 6-7 days/week
- Millets – 4-5 days/week

Fats

- Indian cooking- Use cold pressed Peanut oil, Sesame oil, Ghee.
- Salads and stir fry- Extra Virgin Olive Oil

Proteins

- Legumes-Any variety. 5-6 days/week Soy bean, Rajma, black eyed peas -1 day/week
- Green moong dal sprouts- once week
- Meat products- Eggs 4-5 days/week Chicken/Red meat once a week Seafood once or twice a week

Vegetables

- Include all varieties of vegetables. 4 days/week
- Prefer- Beans, carrots, beetroot, Gourds
- Dark green leafy vegetables like Spinach, Mustard greens. 5 days/week
- Avoid monotonous meals

Dairy

- Prefer buttermilk- 2 cups daily. 6-7 days/week
- Avoid fresh cheeses, butter
- Milk based beverages (Tea, Coffee)- not more than 1 cup a day

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NUTRITION RECOMMENDATIONS

Fruits

- Include all seasonal fruits. 5 days/week
- Avoid fruit juices, instead eat the whole fruit.

Spices

- Reduce salt and chili powder. Rest of the spices okay.

Nuts

- Include chia, flaxseed powder, sunflower seeds, almonds, walnuts. 5 days/week
- Only take one type of nuts/ day About 50 gm/ day

Beverages

- Prefer Green Tea
- Water intake- 1.5 to 2 liter/day

OTHER NUTRITIONAL RECOMMENDATIONS

- Strongly recommend avoiding outside food, especially fried foods- Can cause inflammation.
- Recommend Vitamin E, B complex supplementation.

APPENDIX
Test description &Methodology

Variants associated with the reported clinical condition are analyzed by Whole Exome Sequencing (WES) using the Next Generation Sequence (NGS) analyzer – Illumina's NovaSeq 6000. The DNA libraries were prepared using Roche KAPA hyper Exome panel and sequenced to a mean depth of >80-100x coverage. The panel works on a principle of probe hybridization using DNA probes of around 120mer length providing high and specific interactions with target DNA molecules along with- dual molecular barcodes with dual sample indexing for library preparation. The usage of these dual molecular barcodes reduces or eliminates the index hopping and false positives thus making it more reliable. The NGS test is performed with >95% coverage at a higher depth. Recommended read length of 2 x 150 bps is considered for WES panel sequencing and paired end reads are considered for analysis. Huge quantities of NGS based omics data is concised for better clinical guidance. The raw data with Q30≥80% is considered for analysis, checked for quality by fastQC and aligned to the human reference genome GRCh38.

Variant identification and interpretation are critical steps in making genetic diagnosis and personalized medicine a reality. GATK guidelines are followed for variant identification. The variants were annotated and filtered using the in-house and commercial analysis workflows. Interpretation of sequence variants strictly adheres to the latest ACMG guidelines. In the assessment of the variant classification, GenepoweRx considers information and evidence that includes, but is not limited to, the following 5 significant parameters. The functional impact of the gene in causing the disease phenotype, functional impact of variation in the gene product based on in silico, in vitro, and in vivo studies, variant-disease association, prevalence and significance. This includes comparison against the gnomAD population catalog of variants in 123,136 exomes, the 1000 Genomes Project Consortium's publication of 2,500 genomes, the NCBI ClinVar database of clinical assertions on variant's pathogenicity and multiple lines of computational evidence on conservation and functional impact. The test results are then carefully reviewed and manually curated by our team of highly trained and experienced genome analysts.

Variant Assessment Process

The following databases and algorithms are used to annotate and evaluate the impact of the variant in the context of human disease: 1000 genomes, gnomAD, ClinVar, OMIM, dbSNP, NCIB RefSeq Genes, ExAC Gene Constraints, VS-SIFT, VS-PolyPhen2, PhyloP, GERP++, GeneSplicer, MaxEntScan, NNSplice, PWM Splice Predictor. Analysis was reported using HGVS nomenclature (www.hgvs.org/mutnomen) as implemented by the VarSeq transcript annotation algorithm. The reported transcript matches that used most frequently by the clinical labs submitting to ClinVar.

DISCLAIMER

1. Genetic testing using the methods applied at K&H is expected to be highly accurate.
2. Normal findings do not rule out the diagnosis of a genetic disorder since some genetic abnormalities may be undetectable with this test.
3. This test cannot reliably detect mosaicism. Some genes have inherent sequence properties (for example: repeat, homology, or pseudogene regions, high GC content, rare polymorphisms) that may result in suboptimal data, and variants in those regions may not be reliably identified.
4. This report and the usefulness of the information provided in the report shall not warranty or hold any liability or responsibility for any direct, indirect, incidental, consequential indemnities arising out of the use of, or inability to use the information. This doesn't hold any claims for medico-legal reasons also.
5. This report makes no promises or guarantees that the reported condition/s would develop anytime. Other genetic, environmental, and clinical factors might influence the patients' phenotypic response to the condition. Normal findings do not rule out the diagnosis of a genetic disorder since some genetic abnormalities may be undetectable with this test.
6. False negative results may also occur in the setting suboptimal DNA quality. The chance of a false positive or false negative result due to laboratory errors incurred during any phase of testing cannot be completely excluded. The mutations are not validated/reflex tested.
7. Interpretations are made with the assumption that any information provided on family relationships is accurate. Consultation with a genetics professional is recommended for better interpretation of results. Interpretation of this result should be done in context of a patient's medical record, family history and biochemical profile. Please note that interpretation and classification of the variants reported here may change overtime.
8. Collection, processing, use, storage and retention of the anonymized data, the sample collected and related anonymized reports from the tests conducted for ongoing test developments, educational, scientific research and/or other related activities.
9. Reflex testing by MLPA or multiplex PCR is recommended to confirm the presence of reported CNVs.
10. Please note that only a few key biomarkers are mentioned here, although the analysis was conducted on all biomarkers relevant to the condition of interest. The complete list is available upon request.

This test was developed, and its performance characteristics were determined by GenepoweRx. This test has not been cleared or approved by the U.S. Food and Drug Administration (FDA). The FDA has determined that such clearance or approval is not necessary. These interpretations are based on ACMG/AMP GATK, CPIC Guidelines. This test can be used for clinical screening and research purposes only. The test is registered in CDSCO, India, as Uppalu-Hyder-TE/M/IVD/008407, Class C of Medical Devices, 2022.

Report content approved by:

Dr. Kalyani Palasamudram
Ph.D (Medical Biotechnology)
Chief Scientific Officer

Dr. Kalyan Ram Uppaluri
MD FACP, MD in Internal Medicine
Managing Director

Disclaimer of Liability

This report is provided as an information source for clinicians and is not intended it to be considered a substitute for professional medical advice. If you have or suspect you have a medical problem, contact your physician for personalized treatment or therapy. The translation of genomics knowledge and data into the current report requires generalization and continuous expansion of genomic insights from literature. Hence, minor errors are anticipated. This genomic report and the usefulness of the information provided in the report shall not warranty or hold any liability or responsibility for any direct, indirect, incidental, or consequential indemnities arising out of the use of or inability to use the information. This report makes no promises or guarantees that the reported condition/s will develop at any time. Other genetic, environmental, and clinical factors might influence the patients' phenotypic response to the condition. Proper understanding of the risks aids in better management of metabolic conditions with traditional therapies or other treatment options and helps in the prevention/delay of the disease or makes it less harmful.

Genetic testing plays a key role in the diagnosis of the root cause of a disease or condition. It provides excellent guidance in deciding the right medical regimen. But sometimes, a few non-treatable variants are also identified. Not all genetic changes affect health. It is difficult to know whether identified variants are involved in the condition of interest. Sometimes, an identified variant is associated with a different genetic disorder that has not yet been diagnosed (these are called incidental or secondary findings). A finding of biomarker alteration does not necessarily indicate the pharmacologic effectiveness (or lack thereof) of any drug or treatment regimen; a finding of no biomarker alteration does not necessarily indicate the lack of pharmacologic effectiveness (or effectiveness) of any drug or treatment. No Guarantee of Clinical Benefit: This Report makes no promises or guarantees that a drug will be effective in the treatment of disease in any patient. This Report also makes no promises or guarantees that a drug with a potential lack of clinical benefit will, in fact, provide no clinical benefit. It is possible that a pathogenic variant is present in a gene that was not selected for analysis and /or interpretation in cases where insufficient phenotypic information is available. Due to inherent technology limitations and constant upgradation of research and literature, not all bases of the exome can be covered. Accordingly, variants in regions of insufficient coverage may not be identified and/or interpreted. Therefore, it is possible that pathogenic variants are present in one or more of the genes analyzed but have not been detected. The variants not detected by the assay that was performed may impact the phenotype. For in vitro research use only. This test must be ordered by a qualified medical professional in accordance with required medical regulations.

Patient care treatment decisions must be based on the self-determining medical judgment of the respective physician. Do consider complete patient information such as patient preferences, medical history and family history, physical examination profiles, and other lab results per the standard of care medical practice. The reported results are for the information of the referring doctor only. It should be noted that this test is restricted to a limited number of genes and does not include all intronic and non-coding regions. This report only includes variants that meet a level of evidence threshold for cause or contribute to disease. More evidence for disease association of genes and causal pathogenic variants is discovered every year, and it is recommended that genetic variants are re-interpreted with updated software and annotations periodically.

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